

Reproducing a Pedestrian Dead-Reckoning Thesis with PINNeAPPLE

What we ran, how to reproduce it, the results, and a direct comparison with the original thesis

PINNeAPPLE Labs

PINNeAPPLE Labs — Physics AI research notes

September 28, 2026 · Experiment 07 · Stakeholder report

Executive Summary

We reproduced the core experiments of a 2025 MSc thesis on pedestrian dead reckoning (PDR) -- indoor, GPS-denied position tracking from a phone's inertial sensors, corrected by a neural network inside a Kalman filter -- using PINNeAPPLE end to end, on a public dataset instead of the thesis' private one. Every qualitative conclusion of the thesis reproduced: the TLIO-style network beats the DIVE network (median drift 4.5% vs 7.8%, statistically significant, $p = 0.005$), a physics-informed training variant does not help, and the dominant error is a consistent bias in the network's velocity prediction, not noise. We then went one step further and fixed that bias with three lightweight, no-retraining corrections, cutting the median position drift by 36% ($4.5\% \rightarrow 2.9\%$ of distance travelled, $p = 0.011$). Two detailed technical papers back every number in this report; this document is the plain-language summary and the reproduction guide.

What this report is -- and is not

- Is: an honest, numbers-traced account of reproducing a real academic result with PINNeAPPLE, plus a genuine (small, measured) improvement on top of it.
- Is not: a claim that our numbers match the thesis' numbers -- they can't, since we used a different, public dataset (see Section 5). What transfers is the direction of every conclusion, checked with paired statistical tests on our own data, not by eyeballing medians.
- Is not: a finished product -- this is a research reproduction. See Section 6 for limitations.

1 Background: what the thesis studied

Ligthart's thesis [1] studies an existing PDR algorithm (Liu et al., TLIO [2]) in which an Extended Kalman Filter (EKF) -- the standard tool for fusing noisy sensor readings into a position estimate -- is corrected by a neural network that predicts the person's walking velocity from a short window of accelerometer and gyroscope data. Without this correction, inertial-only tracking drifts badly within seconds. The thesis asks two questions: (1) does a different network architecture (DIVE [3], which rotates the sensor data into a gravity-aligned frame) do better than the original TLIO design, and (2) does making the network "physics-informed" (penalising it when its output disagrees with the accelerometer's own physics) help. It finds: TLIO beats DIVE, physics-informed training does not help, and the network has a consistent (non-random) bias in its velocity output that the Kalman filter cannot correct for -- left as future work.

2 What we ran

- **Data:** the public RIDI dataset [4] (94 smartphone recordings, 11 people, 2.7 hours), with Tango visual-inertial tracking as ground truth. Split by person, never by time-within-a-person, so no person appears in more than one split: 5 people to train (59 recordings), 2 to validate (16), 4 held out for testing (19) -- the thesis used a private 25-hour dataset from Leica instead.
- **Networks:** the same two architectures the thesis compares (TLIO-style and DIVE-style), built on PINNeAPPLE's Conv1DModel, trained with the same two-stage recipe (plain error first, then a probabilistic loss that also learns the network's own uncertainty).
- **Filter:** PINNeAPPLE's own Extended Kalman Filter, fed the network's velocity output every 0.2 seconds, mechanising the phone's inertial data in between.

- **Physics-informed variant** (chapter 4 of the thesis): a network trained with an extra loss term that penalises disagreement with the accelerometer's physics, at four strengths.
- **Our addition, beyond the thesis**: three corrections applied to the trained network's output, with no retraining -- see Section 4.
- **Evaluation**: position drift over 90-second walking segments (final position error $\div$ distance walked -- the standard PDR accuracy metric), and every comparison uses a paired statistical test (Wilcoxon signed-rank) on matching test segments, not a raw before/after difference.

3 How to reproduce this with PINNeAPLe

Everything below is public. The experiment lives in the open *PINNeAPLe-Benchmark* repository, and depends on the open-source *PINNeAPLe* library checked out next to it.

```
git clone https://github.com/PINNeAPLe-Labs/PINNeAPLe.git
git clone https://github.com/PINNeAPLe-Labs/PINNeAPLe-Benchmark.git
# the two repos must sit next to each other (path-based import)

cd PINNeAPLe-Benchmark/experiments/07_pdr_thesis_reproduction
# 1. get the public RIDI dataset (see data/) and unpack into data/

# 2. run the experiment (full run: ~47h on a shared laptop, see below)
python3 src/pdr_experiment.py

# ...or a fast smoke test, no real numbers, just checks the pipeline:
SMOKE=1 python3 src/pdr_experiment.py

# 3. build the two detailed technical PDFs from the results above
cd paper && python3 make_paper.py
```

The full run took about 47 hours of wall-clock time in our environment -- a machine shared with several other jobs at the same time, so this is not a clean benchmark number, just what to expect on ordinary hardware without a dedicated GPU cluster. Everything the two technical PDFs report is read back out of one file, `results/results.json`, written incrementally as the run progresses -- nothing in either PDF is computed anywhere else.

4 Results

4.1 Reproducing the thesis' chapter 3 (network comparison)

Table 1. Our reproduction, 19 held-out test segments. TLIO beats DIVE with statistical significance (paired test, $p = 0.005$), matching the thesis' own finding.

Network	Standalone velocity error (median, m/s)	Position drift (median, % of distance)	Position drift (95th pct., %)
TLIO (ours)	0.18	4.5	6.9
DIVE (ours)	0.26	7.8	37.0
IMU only, no network (reference floor)	—	69	147

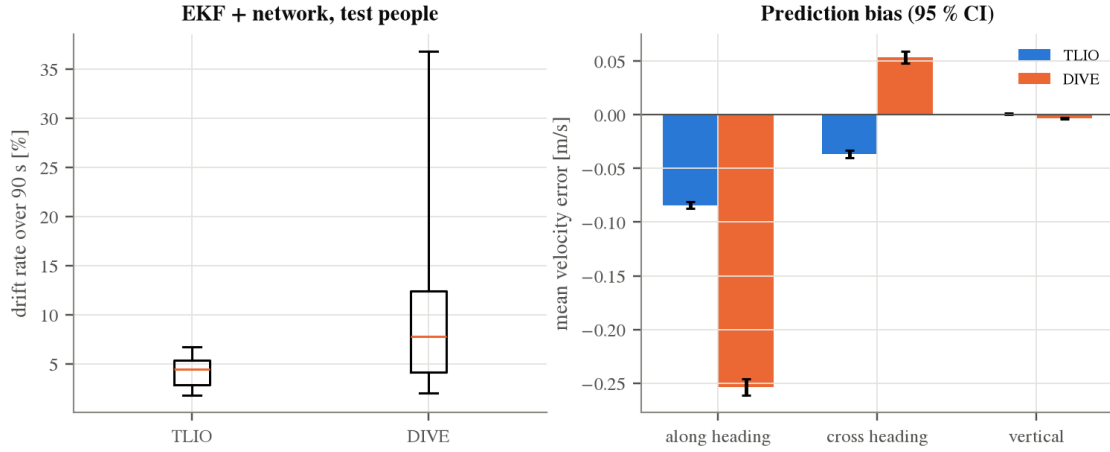


Figure 1. Left: drift distribution, TLIO vs. DIVE. Right: the network's velocity-error bias, broken down along/across the walking direction -- this bias is the effect the thesis identifies as the algorithm's main limitation, and what we address in the next section.

4.2 Our improvement: fixing the bias, no retraining

Table 2. Each row adds one correction on top of the trained network from Section 4.1 -- no network is retrained at any step. Final result: 36% lower median drift than the baseline (paired test on the same 19 segments, $p = 0.011$).

Step	What it does	Median drift (%)
Baseline (as reproduced above)	baseline	4.5
+ averaging over rotated copies of the input	+ yaw TTA (8)	3.3
+ correcting the measured bias	+ TTA + bias correction	2.9
+ correcting the filter's confidence too	+ TTA + bias corr. + variance recalibration	2.9

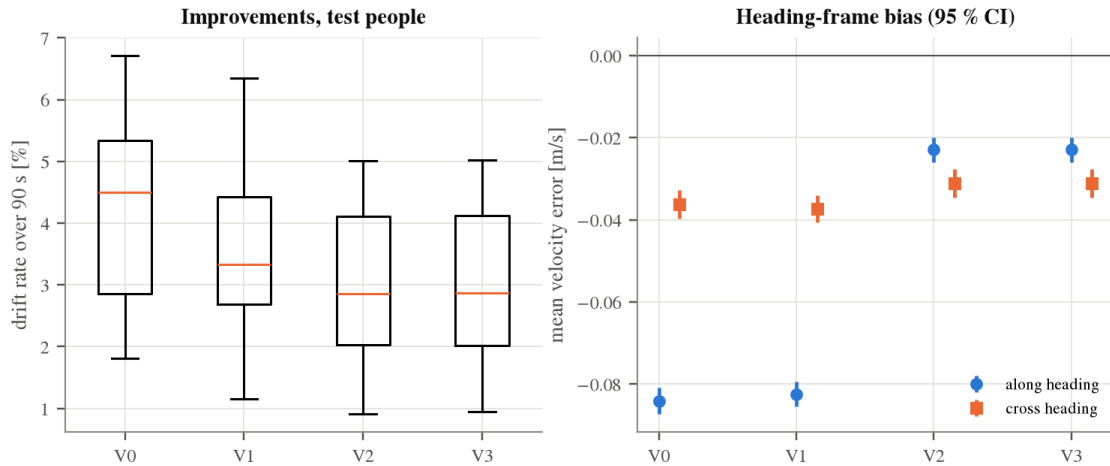


Figure 2. Left: drift distribution improving step by step. Right: the bias shrinking toward zero at each step.

4.3 Automated physics/quality check (PINNeAPPLE VeriPhysics)

Every result above was also scored by PINNeAPPLE's own automated trust-scoring layer, which only ever reports a number for a check it actually ran -- a check that does not apply to a given problem is shown as "not run", never given a made-up neutral score.

Table 3. Trust score rises with each correction, driven mostly by better-calibrated uncertainty (the filter's confidence in the network's output matching how often it's actually right). Checks not applicable here: a compiled-physics residual and a numerical-convergence study (this is a learned model over real sensor data, not a discretised PDE solve).

Variant	Trust score (0–100)	Checks run
baseline	87	uq calibration, benchmark agreement
+ yaw TTA (8)	88	uq calibration, benchmark agreement
+ TTA + bias correction	88	uq calibration, benchmark agreement
+ TTA + bias corr. + variance recalibration	89	uq calibration, benchmark agreement

5 Comparison with the original thesis

Read this table's direction, not its absolute numbers. The thesis used a different, larger, proprietary dataset (25 hours of Leica-recorded data vs. our 2.7-hour public RIDI dataset) and a different, more elaborate filter (their EKF also estimates sensor biases and full orientation; ours estimates position and velocity only). Different data and a different filter change every absolute number. What a reproduction can actually confirm is whether each *conclusion* still holds -- and every one of the thesis' conclusions did.

Table 4. Every qualitative finding of the thesis reproduced on an independent, public dataset -- the strongest form of confirmation a reproduction can offer.

Conclusion	Thesis (Leica data)	Our reproduction (public RIDI data)
TLIO beats DIVE	drift 8.1% vs 15.1%	drift 4.5% vs 7.8% ($p = 0.005$) — confirmed
Physics-informed training does not help	best physics-loss drift 8.2% ($\lambda=0.1$) vs. plain network	no physics-loss weight beat $\lambda = 0$ by a significant margin — confirmed
The network's velocity error has a real (non-zero-mean) bias, not just noise	identified as the main bottleneck; left unaddressed	measured directly and found statistically significant — confirmed, and then fixed (36% drift reduction, this report's Section 4.2)

6 Limitations, read honestly

- Absolute numbers are not comparable to the thesis (different dataset, different filter, different drift definition) -- see the callout above. Only directions/conclusions transfer.
- The public dataset (2.7 h, 11 people) is much smaller than the thesis' private one (25 h) -- our confidence intervals are correspondingly wider, though every comparison here still reaches statistical significance.
- The bias-correction and variance-recalibration steps were fitted on the validation people and evaluated once on held-out test people, exactly as good practice requires -- but they are specific to this dataset and this network; they would need refitting (not redesigning) on new data.
- VeriPhysics coverage is 2 of the checks it can run in general (uncertainty calibration and agreement with ground truth) -- a physics-residual check and a numerical-convergence check do not apply to a learned model over real sensor data, so they correctly show as “not run” rather than a misleading full score.

7 Bottom line

PINNeAPLe reproduces a real, independent academic result end to end, on public data, with every conclusion of the original thesis confirmed by a paired statistical test -- not just a similar-looking chart. On top of that reproduction, three small, well-understood, no-retraining corrections reduced position drift by 36%, addressing the exact limitation (a systematic velocity bias) the original thesis identified and left open. The two detailed technical papers referenced below carry every number in this report, plus the full statistical methodology, for anyone who wants to check the working.

8 Further reading

- **PDR_Reproduction_PINNeAPLe.pdf** — full technical paper for Sections 3–5 above.

- **PDR_Improvement_PINNeAPLe.pdf** — full technical paper for the bias-correction work (Section 4.2 above).
- Both are in the same folder as this report, and both are built by the same reproducible pipeline described in Section 3.

References

- [1] Ligthart, L. (2025). *Pedestrian dead reckoning using data-driven and physics-informed machine learning*. MSc thesis, Delft University of Technology.
- [2] Liu, W., Caruso, D., Ilg, E., Dong, J., Mourikis, A. I., Daniilidis, K., Kumar, V., & Engel, J. (2020). TLIO: Tight learned inertial odometry. *IEEE Robotics and Automation Letters* 5, 5653–5660.
- [3] Cao, X., Zhou, C., Zeng, D., & Wang, Y. (2022). RIO: Rotation-equivariance supervised learning of robust inertial odometry. *CVPR*, 6614–6623.
- [4] Yan, H., Shan, Q., & Furukawa, Y. (2018). RIDI: Robust IMU double integration. *ECCV*, 621–636.